



A SYSTEM AND METHOD FOR CLEANING A SOLAR PANEL

✓ KEY INFORMATION

Patent Number: PI 2024000995

Technology Readiness Level: TRL 8

Market Readiness Assessment: MR 7

Location: Universiti Malaya, Malaysia



✓ TECHNOLOGY OVERVIEW

Traditional solar panel cleaning methods are manual, labor-intensive, and pose safety risks. Our **Automated Solar Panel Cleaning System** is an innovative, efficient, and safe solution that uses smart technology to maintain optimal solar panel performance.

The system features **smart sensors**, a **curved-flow fluid distribution panel**, and **automated control** to clean solar panels without high-pressure water, reducing damage risks and operational costs. It is designed to be **scalable, modular, and compatible** with various solar panel types and installations.

Through real-world testing, our system has demonstrated the ability to **restore 15–30% of lost energy output**, providing a **sustainable and cost-effective solution** for solar energy systems.

✓ FEATURES AND SPECIFICATIONS

Our **Automated Solar Panel Cleaning System** is designed to maximize cleaning efficiency while minimizing water usage and maintenance. Key features include:

- **Smart Fluid Distribution Panel**
Utilizes a curved-flow design to evenly distribute cleaning fluid **without high pressure**, suitable for panel angles from 0–90 degrees.

- **Automated Control System**
AI-driven processing device monitors panel performance and initiates cleaning cycles when efficiency drops below a set threshold.
- **Multi-Agent Integration**
Supports water, cleaning agents (5–25 mL/L), and protective coatings to enhance panel performance and longevity.
- **Sensor Network**
Light sensors (photodiodes, photoresistors) detect dust accumulation and distinguish it from weather-related performance drops.
- **Low Maintenance and Long Lifespan**
Designed for durability with minimal operational costs, reducing maintenance expenses by up to **40%** compared to manual cleaning.

✓ POTENTIAL APPLICATION

This **automated cleaning technology** is suitable for various solar energy applications:

- **Utility-Scale Solar Farms** – Provides efficient, large-scale cleaning for solar parks.
- **Commercial Rooftop Installations** – Safe and automated cleaning for urban solar arrays.
- **Off-Grid Solar Systems** – Ideal for remote locations with limited maintenance access.
- **High-Dust Environments** – Effective in deserts and industrial areas with heavy particulate accumulation.
- **Water-Scarce Regions** – Optional coating application reduces water dependency.

✓ UNIQUE VALUE PROPOSITIONS

Our **Automated Solar Panel Cleaning System** offers a transformative approach to solar panel maintenance, outperforming conventional methods in efficiency and safety:

Cost-Effective and Efficient

- Reduces manual labor costs by **50–70%**
- Restores **15–30%** of lost energy output
- **40%** lower maintenance costs compared to traditional methods

Safe and Eco-Friendly

- Eliminates manual cleaning risks (e.g., falls, electrocution)
- Uses **30% less water** than manual methods and supports water recycling

Smart and Adaptive

- AI-driven system optimizes cleaning cycles based on real-time data
- Compatible with various solar panel types and installation angles (0–90°)

Proven Performance

- Successfully tested in pilot projects, ready for commercial deployment
- Developed at **Universiti Malaya** in collaboration with industry partners

YOU HAVE INQUIRY? >>>

For more information, contact:
Dr. Lee Ching Shya (RTTP) | leecs@um.edu.my
Dr. Nurul Fauzani Jamaluddin | nfauzani@um.edu.my
UM Centre of Innovation and Enterprise (UMCIE)
umcie@um.edu.my
Tel: +603-79677351